Random Variables Probability Distributions

$$E(X)= \sum xP(X = x)$$

Number of Sales	Probability
0	0.2
1	0.5
2	0.3

What does the expected number of sales represent?

E(X) = 0(0.2)+1(0.5)+2(0.3)

E(X) 1.1

Expected value = long run average outcome, not necessarily an outcome that must actually occur.

VARIANCE OF A RANDOM VARIABLE

Variance of a random variable measures the expected squared deviation of the random variable from its expected value.

For Discrete variable:

$Var(X) = E[(X-\mu)^2]$

where:

X = possible outcome

μ = (E(X) = Expected value (Mean)

P(X = x) = probability of outcome

Var(X) = variance

	Marks	(X- u) DEVIATI	(X-u)^2		
	60	1.7	2.777778		
	50	-8.3	69.44444		
	55	-3.3	11.11111		
	60	1.7	2.777778		
	45	-13.3	177.7778		
	80	21.7	469.4444		
			733.3333		
Average(Mear	58.33333				
Variance			122.2222	SD	11.05542

Variance tells us how widely the possible outcomes are spread around the expected value.

Two stores may have the same expected daily sales\				Expected(Mean Sales value)
Store A	90	100	110	100
Store B	20	100	180	100

Both can have an average around 100, but Store B has much greater variability.

Same expected value <> same level of risk

Variance formula:

$$Var(X) = \sum[(X-\mu)^2P(X=x)]$$

$(X-\mu)$-> deviation

$(X-\mu)^2$ --> Square the deviation

Multiply the probability:

$(X-\mu)^2P(X=x)$

Add everything

Random Variables Probability Distributions

X	P(X)	(X-μ)	$(X-\mu)^2$	$(X-\mu)^2P(X=x)$
0	0.2	-1.1	1.21	0.242
1	0.5	-0.1	0.01	0.005
2	0.3	0.9	0.81	0.243
			Var(X)	0.49
	E(X) μ	1.1		

Interpretation:
The variance measures the spreadn of the possible number of sales around the expected value of 1.1

STANDARD DEVIATION

Standard deviation is the postive square root of the variance of a random variable

σ=(Var(X))^(1/2)

σ = 0.7

Standard deviation tells us the typical amount by which outcomes vary around the expected value

Why we need SD if variance already exists?
Variance is expressed in squared units
If sales are measured in:
units^2
that is difficult to interpret

Standard deviation returns to the original unit.
Example:
Expected sales = 1.1 orders
Variance = 0.5 orders^2
SD = 0.7 orders
So SD is easy to interpret.

BUSINESS APPLICATIONS
Why Variance & SD Matters in business?
Example: Customer orders
Suppose two products have the same expected daily demand.
Product A Demand remains stable
Product B Demand fluctuates heavily
Even if **E(A) = E(B)**
their risk is different.

Random Variables Probability Distributions

Applications

1. Inventory

Higher demand variability --> greater uncertainty in stock requirements

2. Finance

Higher variabiliy in returns --> Greater uncertainity

3. Operations

Higher variation in call volume --> staffing becomes harder.

4. Marketing

Variation in customer spending can help identify inconsistent purchasing patterns.

How much does the outcome vary around the average?

X	P(X)	(X-μ)	$(X-\mu)^2$	$(X-\mu)^2P(X=x)$
1	0.2	-1.1	1.21	0.242
2	0.5	-0.1	0.01	0.005
3	0.3	0.9	0.81	0.243
			Var(X)	0.49

Expected value	2.1
Variance	0.49
Standard deviation	0.70

BINOMIAL DISTRIBUTION

It is a discrete probability distribution that models the number of successes in a fixed number of independent trials, where each trial has two possible outcomes and the probability of success remains constant.

Binomial distribution answers: " Out of a fixed number of attempts, how many times will success happen?"

Conditions of binomial distribution

BINS idea

B--> Binary outcomes

Only two outcomes

1. Success/Failure
2. Yes/No
3. Defective/Non-Defective
4. Purchased/Non-Purchased

I--> Independent trials

One trial does not affect another.

N--> Number of trials is fixed

Example: 10 customers.

not an unlimited number.

Random Variables Probability Distributions

S--> Same probability of success

Probability remains constant

Example: p= 0.2

for every trial.

Binomial Formula

$$P(X=x) = \binom{n}{x} p^x (1-p)^{(n-x)}$$

n= total number of trial

x= number of success

p= probability of success

(1-p)= probability of failure

$\binom{n}{x}$ = number of ways to arrange the success (Cobinations)

Example:

A salesperson contact 5 customers

Probability that a customer purchases:

p= 0.2

What is the probability of exactly 2 customers purchasing?

Then

n = 5

x = 2

p = 0.2

(1-p) = 0.8

Therefore

P(X = 2) =(5C2)(0.2)^2(0.8)^3

0.2048

P(X = 2) = 20.48%

nCr =n!/r!(n-r)!

r!(n-r)! = 12

n! = 120

10

n!	120
r!	2
(n-r)!	6

0.2048

FALSE --> exactly 2 successes

0.94208

TRUE--> Probabilty of 2 or fewer

sales	P(X)	
1	0.2	P(exactly 3)
2	0.1	P(at least 3)
3	0.3	P(atmost 3)
4	0.4	P(2 or fewer)

Random Variables Probability Distributions

INTERVIEW

Q. What are the conditions of binomial distribution?

BINS

Fixed number of trials

Two possible outcomes

Independent trials

Constant probability of success

What does FALSE mean in BINOM.DIST()?

It calculates the probability of exactly x success.

What does TRUE means?

It calculates the cumulative probabilty, meaning the probability of x or fewer successes(at most x success)